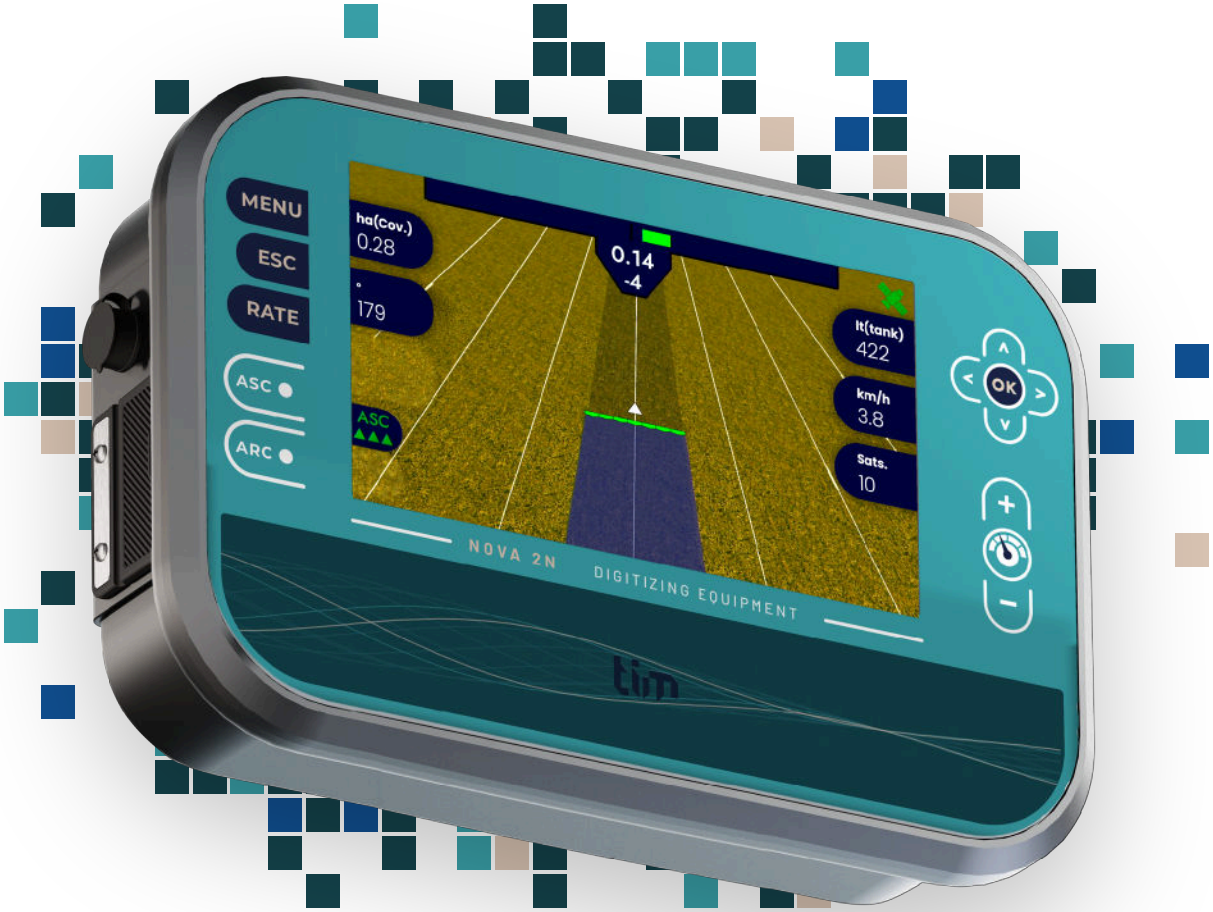
DRAFT · V.0.2



NOVA 2N

GNSS Navigation Computer

Installation & User Manual

TNOV2N.01 · GNSS Navigation

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About this Document

Document Information

Document Type	Installation & User Manual
Document Abstract	Overview and detailed instructions for the installation and use of the NOVA 2N GNSS navigation device
Product	NOVA 2N (TNOV2N.01)
Software	The installed version is shown under Data Display → Device Data
Document Status	V.0.2 – **DRAFT** (items marked *To be confirmed* are pending R&D verification)
Document Prepared By	Timdigitize Documentation Team

Document Revision History

Sr.	Version	Date	Comments
1.	0.1	09-Oct-2026	First draft
2.	0.2	09-Oct-2026	Navigation only: no implement selection, no job storage, sprayer kits removed
3.			

This manual is an integral part of the product and must accompany it in case of sale or change of ownership. Keep it for future reference. Screen illustrations are taken from the NOVA 200 platform on which NOVA 2N is based; the screens of your device may differ slightly. Screen texts are written **like this**. Tim reserves the right to modify product specifications and instructions at any time without notice.

1. Symbols & Abbreviations

The installation tasks described in this manual must be performed by qualified technicians only.

1.1 Symbols Used

Symbol	Type of Hazard	Description
	**DANGER**	Indicates a hazardous situation that needs immediate attention. If not avoided, serious injury or even death may occur.
	**WARNING**	Indicates a hazardous situation that could lead to death or serious injury if precautions are not taken.
	**CAUTION**	Indicates a situation that might result in minor injury or damage to the product if not avoided.
	**NOTE**	Provides tips that are valuable for the optimal operation of your product.

1.2 Abbreviations

Term	Meaning
GNSS	Satellite positioning (GPS, Galileo, BeiDou ...). NOVA 2N uses the Basic GNSS antenna, the Cross 200 Smart GNSS or the Cross 300 Smart Antenna
A–B line	Straight guidance line defined by two points A and B
Parallel lines	Guidance lines next to the A–B line, at the line spacing
Line spacing	Distance between two neighbouring guidance lines – normally the working width of the implement you use
Lightbar	Bar at the top of the screen that shows how far the tractor is off the guidance line
Shift line	Moves the current guidance line to the actual tractor position
Pause point	Marked position (e.g. before refilling) to which NOVA 2N guides you back
Boundary	Field perimeter recorded with GNSS
Coverage	Area already driven, painted on the map; overlaps and gaps become visible
HDOP	Horizontal dilution of precision – quality of the GNSS position (lower is better)

2. Before Using the Product

ℹ	Carefully read all the safety instructions. Otherwise, there is a risk of accident, injury or damage to the product.
❗	Never clean the device or the antenna with direct high-pressure water jets.
❗	Do not use solvents or fuel to clean the outer surfaces.
⚠	Comply with the specified supply voltage: **12 VDC**. Reverse polarity can cause irreversible damage.
⚠	Do not expose the device to temperatures above 70 °C. Do not store or place flammable substances near the product or its wiring.
⚠	Before any electric arc welding on the tractor, disconnect all cables from NOVA 2N.
⚠	Watch the field, not only the screen. NOVA 2N does not steer the tractor; the driver remains responsible for safe driving at all times.
ℹ	Only use genuine Tim cables, spare parts and accessories.

2.1 Intended Use

NOVA 2N is a GNSS navigation device for the tractor cab. It shows the driver how far the tractor is off a straight guidance line, so that passes are driven straight and parallel at a constant spacing – for spraying, spreading, seeding, mowing or any other field work.

NOVA 2N does **not** control the implement (no section or rate control, no implement selection) and does **not** steer the tractor. Any other use is considered improper. Tim is not liable for damage resulting from improper use or failure to observe this manual.

2.2 Important Safety Instructions

- Connect NOVA 2N only to a fused 12 VDC supply using the supplied power cable.
- Do not route cables near hot parts, sharp edges or the PTO shaft.
- Mount the device so that it does not block the driver's view and cannot come loose while driving.
- Do not operate the menus while driving on public roads.
- Do not open the housing. There are no user-serviceable parts inside.
- Electronic components are sensitive to electrostatic discharge (ESD). Touch a grounded metal part before handling connectors.

2.3 Installation Pre-requisites

1. The tractor is switched off and parked safely on level ground.
2. There is no excessive humidity, dust or dirt, and no flammables or explosives nearby.
3. All federal, state and local site safety requirements are in place.

3. About the Product

3.1 About NOVA 2N

NOVA 2N is a simple GNSS navigation device on the NOVA 200 platform. You mark a straight A–B line once; NOVA 2N then draws parallel lines at the line spacing and the lightbar shows how far the tractor is off the line. No implement has to be set up – the device only shows the tractor on its straight line.

- **Parallel A–B guidance** with lightbar.
- **Shift line** – move the line to the tractor position when it has drifted.
- **Pause point** – find the place where you stopped, e.g. after refilling.
- **Field boundary** recorded with GNSS.
- **Coverage map** – the area already driven is painted on the map and counted in hectares.

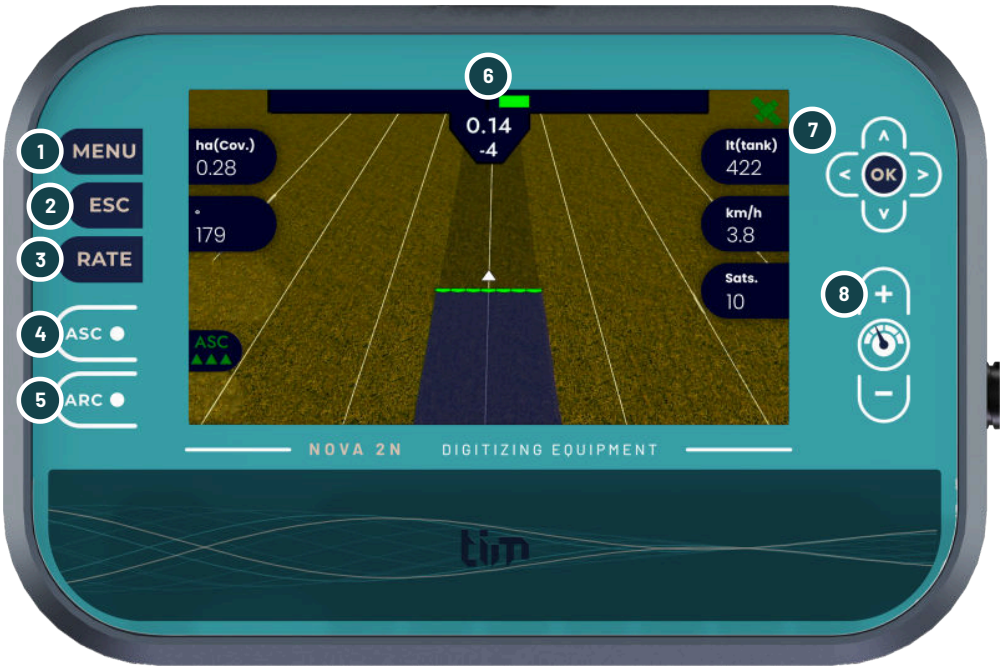
3.2 Specifications

Specification	Value
Display	7" colour display, 1024 × 600, day / night mode
Supply voltage	12 VDC
GNSS	Basic GNSS antenna (KYP028, standard) · Cross 200 Smart GNSS (KYP036) · Cross 300 Smart Antenna (KYP067)
Front panel	Keys only, no switches
Data	microSD card (boundaries), Wi-Fi (software update)
Languages	English, Русский, Türkçe, Ελληνικά, Español, Português

? **TO BE CONFIRMED** · Specifications are taken from the NOVA 200 platform. Operating temperature, protection class, dimensions and weight of NOVA 2N are to be confirmed.

3.3 Product Overview

Note: The final product may differ from the illustration depending on the production batch and customer specifications.



No.	Part	Function
1	MENU	Back to the HOME menu.
2	ESC	Back / cancel. On the run screen: switches the A–B guidance on / off and shows the active alarms.
3	RATE	Not used for navigation (common front panel of the NOVA 200 platform).
4	ASC + LED	
5	ARC + LED	
6	Display	7" colour display.
7	Navigation keys + OK	Menu navigation; on the run screen ∧ ∨ zoom, < opens the left panel, > the right panel.
8	+ / –	Not used for navigation.

? **TO BE CONFIRMED** · RATE, ASC, ARC and + / – have no navigation function on NOVA 2N – whether they are inactive or have another function is to be confirmed.

4. Installation

The installation must be carried out by a skilled professional only. If the mounting bracket is not securely fastened, the device can move or fall during operation and cause damage or injury.

4.1 System Overview



NOVA 2N needs only three connections: the GNSS antenna on the cab roof, the GNSS harness and the power cable. Nothing is connected to the implement.

4.2 Mounting the Device

Step 1. Unbox the equipment and place NOVA 2N on a flat, clean surface.

Step 2. Mount it with the triple suction cup mount **KYP029** (optional) on a clean, smooth glass or panel surface, or with the screw mount **KYP037** on a rigid part of the cab.

Step 3. Place it within reach of the driver and in the line of sight, out of direct sunlight, without blocking the view.

4.3 GNSS Antenna



KYP028

Mount the GNSS antenna – the Basic GNSS antenna **KYP028** supplied with NOVA 2N, or the optional Cross 200 / Cross 300 – on the **cab roof**, at the highest point, **exactly on the centre line of the tractor**, with a clear view of the sky. Never install the antenna inside the cab. Connect it to NOVA 2N with the Smart GNSS harness **KYP054**.

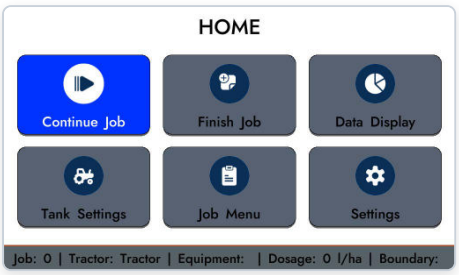
Measure the distance from the antenna to the rear axle with a tape measure – it is entered as GPS antenna / Rear Axle in the tractor profile (section 5.4.3). For Cross 200 / Cross 300 see their own manuals.

4.4 Power Cable

Connect the power cable **SIM014** directly to the tractor battery (pin 1 black = –, pin 2 red = +). Use a fused line; do not share it with high loads.

5. Software Operations

5.1 HOME Menu



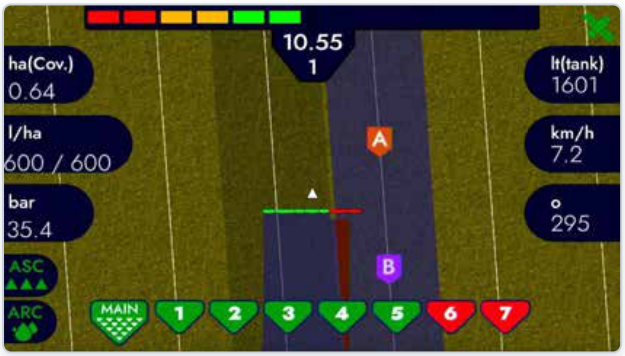
HOME (NOVA 200 platform)

Item	Function
Start New Job	Opens the run screen with an empty map: no A–B line, no coverage.
Data Display	GPS data, device data, field overview.
Settings	Guidance, user, advanced and tractor settings (section 5.4).

On the first start NOVA 2N searches the GNSS receiver (**Searching for GPS sources**) and shows **GPS Connection Successful!** or **GPS Connection Failed!** .

> ? **TO BE CONFIRMED** · The HOME menu of NOVA 2N is a reduced version of the NOVA 200 menu (no job storage, no implement or tank settings). Exact menu items and names are to be confirmed.

5.2 Run Screen

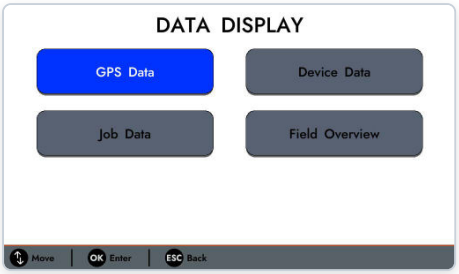


Run screen with A–B guidance (NOVA 200 platform)

- **Lightbar** (top) – deviation from the guidance line (m).
- **Data boxes** – configurable values (User Settings → Configure Display), e.g. ha(Cov.), heading, km/h, satellites.
- **Map** – tractor, A–B and parallel lines, coverage, boundary, pause point.
- **Satellite symbol** – green = GNSS position OK.

Key on the run screen	Function
∧ ∨	Zoom in / out (8 levels)
< left panel	Day / night mode · 3D / 2D view
> right panel	Shift line · pause point · A–B line · field boundary (requires GNSS)
ESC	A–B guidance on / off · alarm list
MENU	HOME

5.3 Data Display



DATA DISPLAY (NOVA 200 platform)

- **GPS Data** – longitude, latitude, data frequency, number of satellites, date and time, HDOP, speed. **OK** changes the GPS baud rate.
- **Device Data** – hardware and software version, CPU temperature, IP address.
- **Field Overview** – the whole field: covered, overlapped and uncovered area. Arrows pan, + / – zoom.

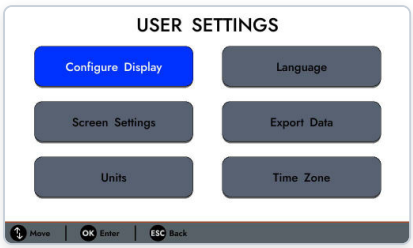
5.4 Settings

5.4.1 Guidance

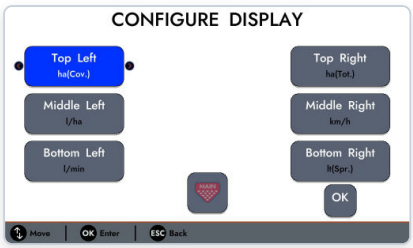
Item	Function	Default
Line spacing	Distance between the guidance lines; also the width of the coverage painted on the map. Enter the working width of the implement you use (e.g. 24 m for a 24 m spreader).	–
Ref. Line Offset	Fine correction of the line distance (**Increase** / **Decrease**). Line distance = line spacing ± offset.	0 m
Lightbar Sensitivity	Deviation per lightbar step.	0.5 m

? **TO BE CONFIRMED** · Where the line spacing is entered on NOVA 2N (menu and name) is to be confirmed.

5.4.2 User Settings



USER SETTINGS



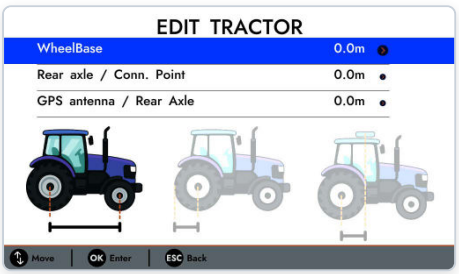
CONFIGURE DISPLAY



LANGUAGE

- **Configure Display** – choose the value of each data box, e.g. ha(Cov.), km/h, heading, satellites. Confirm with **OK**.
- **Screen Settings** – brightness. **Units** – metric. **Time Zone** – UTC –12 … +12 (default +3).
- **Language** – English, Русский, Türkçe, Ελληνικά, Español, Português.

5.4.3 Tractor



EDIT TRACTOR

Create Tractor / **Edit Tractor** – tractor name and geometry. NOVA 2N uses it to place the tractor correctly on the map:

- **Wheelbase:** front axle to rear axle (m).
- **Rear axle / Conn. Point:** rear axle to the hitch (m).
- **GPS antenna / Rear Axle:** antenna on the roof to the rear axle (m).

5.4.4 Advanced Settings

- **Test Menu** – keys light green when pressed; SD card status.
- **Software Update** – requires Wi-Fi with internet: **Checking for updates** → **Press OK to Update** → **Update Complete**. Do not switch off during the update.
- **Factory Settings** – **Factory Reset** deletes all settings, tractor profiles and boundaries; **Screen Default** resets the display settings.
- **WIFI Connection** – select a network, enter the password with the on-screen keyboard, **Connect**.
- **Audio Settings** – button and alarm sound.

6. Operating Instructions

6.1 First Operation

1. Switch NOVA 2N on and wait for the HOME menu and **GPS Connection Successful!** .
2. **Language and time zone**: Settings → User Settings.
3. **Tractor**: Settings → Tractor → **Create Tractor** . Measure all distances with a tape measure.
4. **Line spacing**: enter the working width of your implement (section 5.4.1).

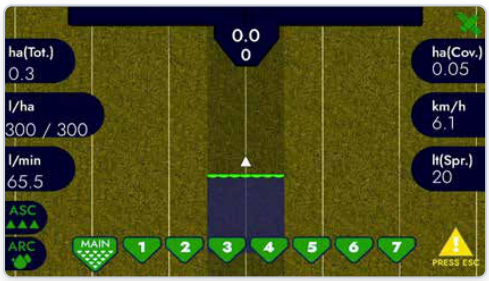
> ⚠ A wrong line spacing, wrong tractor geometry or an antenna that is not on the tractor centre line causes gaps or overlaps between the passes and a shifted coverage map.

6.2 Start Navigation

1. HOME → **Start New Job** . The run screen opens with an empty map.
2. Wait until the satellite symbol is green, then drive to the start of the first pass.

> ℹ NOVA 2N does not store jobs. A new start begins with an empty map; lines and coverage of the previous work are not kept.

6.3 A–B Line and Parallel Lines



A–B guidance (NOVA 200 platform)

1. At the start of the first pass: run screen → **>** → A–B line icon → **OK -> Mark A?** → **OK** .
2. Drive straight; NOVA 2N shows **Move ... m away from A** . After at least **30 m** press **OK** at **Mark B?** – the best result is at the end of the pass.
3. Parallel lines are drawn at the line spacing. At the end of each pass, turn onto the next line and keep the lightbar centred.

- **ESC** switches the A–B guidance on and off. A, B and the pause point are kept after switching off.
- If the passes do not match the implement, correct the line spacing or the **Ref. Line Offset** .

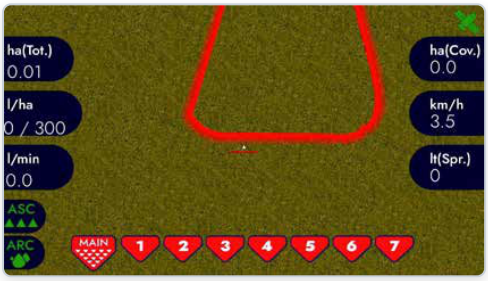
6.4 Shift Line

GNSS positions drift slowly over time, and a field edge is rarely perfectly straight. If the tractor is exactly where it should be but the lightbar shows a deviation, open the right panel (>) and select **shift line**: the guidance line moves to the current tractor position and all parallel lines move with it.

6.5 Pause Point

1. Before leaving the field (refilling, break), open the right panel (>) and set the **pause point**.
2. Drive back: the distance to the pause point is shown on the screen. Continue the pass where you stopped – the coverage map shows what is already done.
3. Clear the pause point in the right panel.

6.6 Field Boundary



Recorded boundary (red line)

1. Run screen → > → boundary icon.
2. **Field Boundary** : choose **Left** or **Right** – the side of the tractor that runs along the edge (Left = drive clockwise, Right = counter-clockwise).
3. **Press OK to start** and drive around the field at steady speed. < pauses / continues; **OK** finishes.
4. Enter a name for the boundary.

Keep a steady speed and a good GNSS signal while recording. An SD card is required (No SD card. Failed!).

? **TO BE CONFIRMED** · Whether NOVA 2N keeps recorded boundaries for later use, and how a saved boundary is selected without the Job Menu, is to be confirmed.

6.7 Coverage Map

While you drive, the area behind the tractor is painted on the map at the line spacing width, and ha(Cov.) counts the covered area. Data Display → **Field Overview** shows the whole field with covered, overlapped and uncovered areas – check it before leaving the field.

? **TO BE CONFIRMED** · Without a main switch: whether coverage is painted all the time while driving or can be started / stopped from the run screen is to be confirmed.

7. Alarms & Troubleshooting

Alarms are shown in a pop-up with a sound (Audio Settings). ESC on the run screen lists the active alarms.

Alarm	Cause	Action
NO GPS SIGNAL!	No GNSS data for more than 5 s	Check the antenna on the cab roof and its cable
ORIGIN FAR AWAY	The field origin is more than 10 km away	Start a new job for this field
NO SD CARD!	No SD card inserted	Insert the SD card (boundaries)

7.1 Troubleshooting

Problem	Solution
GPS Connection Failed!	Check the antenna cable; Data Display → GPS Data → OK to select the baud rate.
Gaps or overlaps between passes	Check the line spacing and Ref. Line Offset against the real working width of the implement.
Lightbar shows a deviation although the tractor is on the line	GNSS drift – use shift line (section 6.4).
Coverage map shifted to one side	Antenna not on the tractor centre line, or wrong tractor geometry.
Position jumps / unstable lightbar	Few satellites or high HDOP – check the sky view of the antenna; for higher precision use Cross 200 or Cross 300.
Boundary cannot be saved	Insert an SD card.

8. Package Contents

8.1 Core Elements

Code	Description	Image
TNOV2N.01	NOVA 2N	
SIM014	2 m Power Cable	
KYP028	Basic GNSS Antenna	
KYP054	Smart GNSS Harness	

8.2 Optional Products

Code	Description	Image
KYP029	Triple Suction Cup Mount · 6 cm short arm	
KYP036	Cross 200 Smart GNSS	
KYP067	Cross 300 Smart Antenna	

9. Disposal



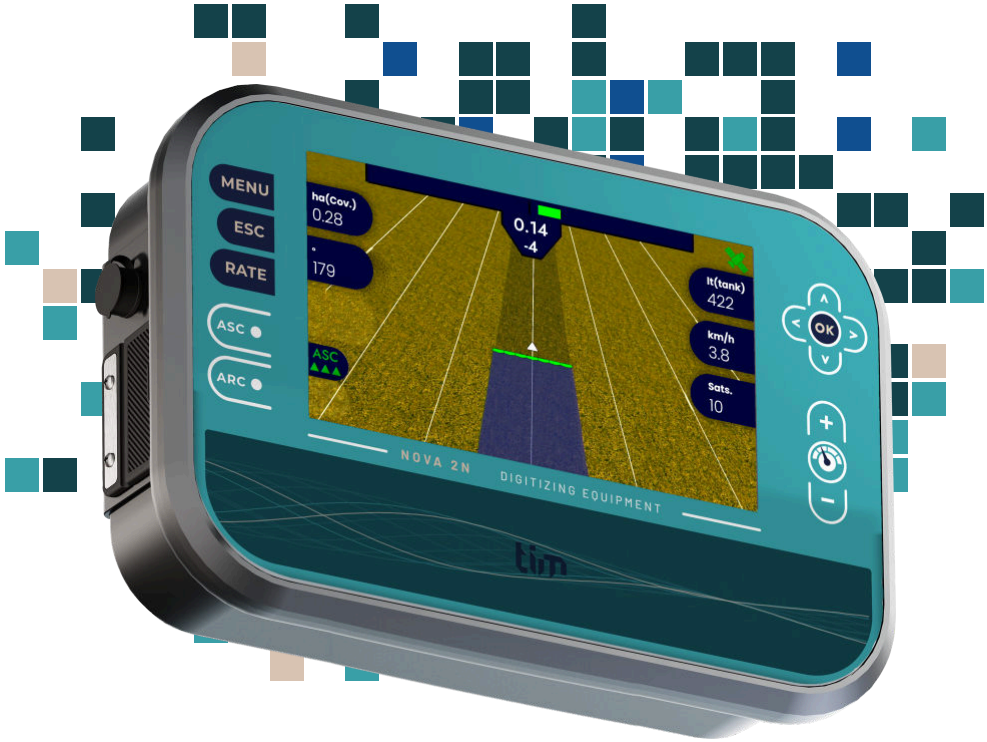
9.1 For business users in the European Union

The crossed-out wheeled bin on the product, its accessories or packaging indicates that this product must not be treated as household waste. Dispose of it at a collection point for the recycling of electrical and electronic equipment (WEEE). Correct disposal saves valuable natural resources and prevents possible negative effects on the environment and human health. Ask your local authorities or your Tim dealer for details. Improper disposal may be subject to fines under national regulations.

10. Warranty

Every Tim product is covered by a **2-year warranty** from the date of purchase. The warranty does not cover damage caused by improper installation or use, use of cables or accessories not supplied by Tim, opening of the housing, accidents, or failure to observe this manual. For warranty claims contact your Tim dealer.

Changes to the text, design and technical specifications may occur without prior notice; we reserve the right to make these changes.



Tim Dijital Teknolojiler A.Ş.
Esenşehir Mah. Haseki Sok. No:20/2
Ümraniye – İstanbul / Türkiye
Tel: +90 216 499 71 90
E-mail: info@timdigitize.com